

social-agent

// autonomous social-video pipeline · v0.3.0

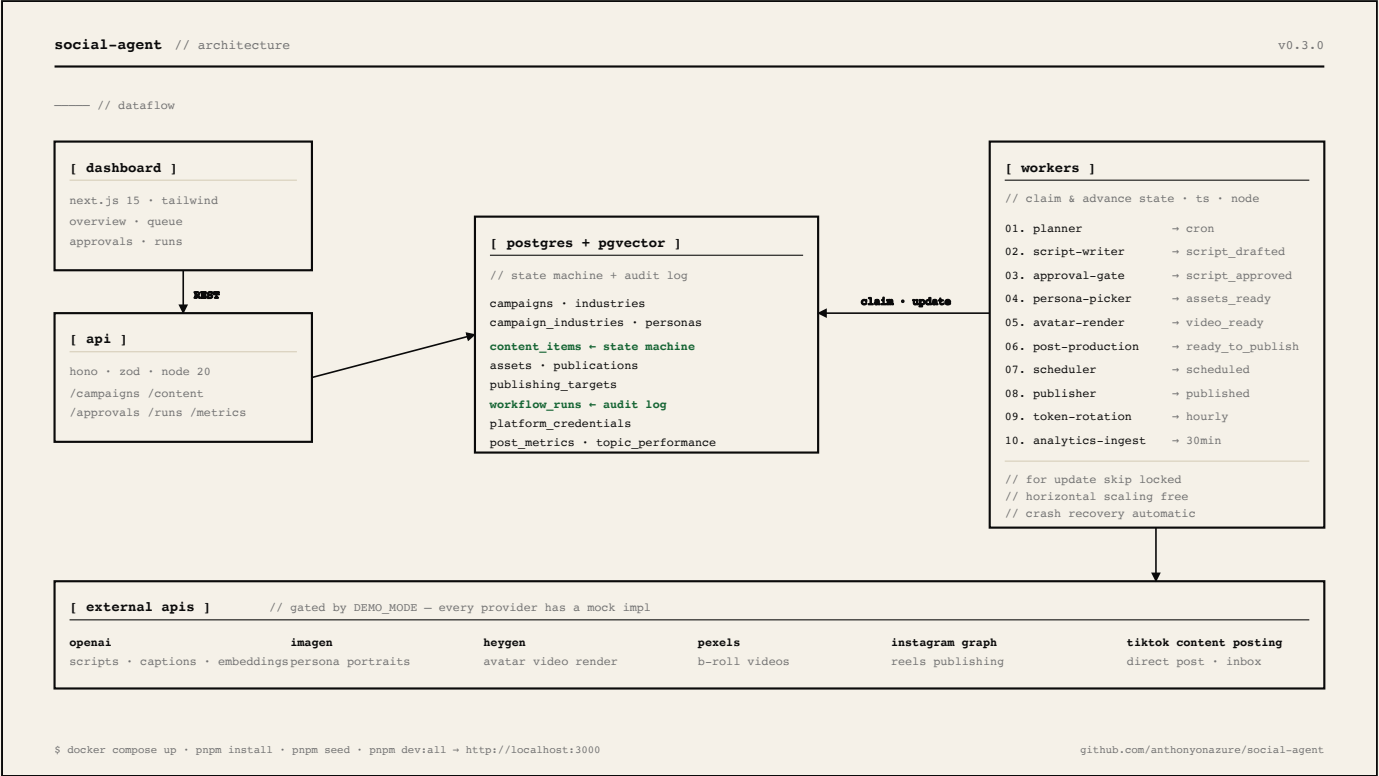
—— // §1 thesis

self-running social-video pipeline.

From campaign config to published Reels and TikToks – autonomously.
State-machine in Postgres, stateless TypeScript workers, n8n
orchestration, live dashboard with HITL approval gates that flip per-
campaign to fully autonomous.

CODE	WORKERS	STATES	DEMO COST
4,000+	11	13	\$0
// lines of typescript	// stateless	// audit logged	// no paid keys

—— // §2 architecture



// \$1 overview

overview

// pipeline health across all campaigns

PLANNED

00

// awaiting script

IN_FLIGHT

03

// workers active

SCHEDULED

06

// publish queue

PUBLISHED

24

// live on platforms

FAILED

00

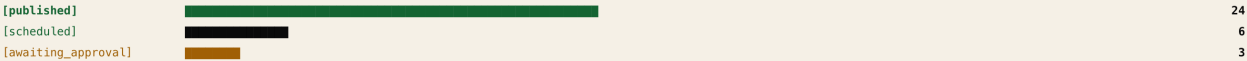
// needs review

// \$2 campaigns

NAME	MODE	CADENCE	PLATFORMS	WEEKLY
Demo Brand └ portfolio demo: a fictional growth-marketing brand serving local businesses across multiple verticals.	[hitl]	0 9,17 * * *	 	33 /wk

manage →

// \$3 state distribution



\$ npm dev:all • 127.0.0.1:3000

[github.com/anthonyonazure/social-agent](#)

social-agent

// case study · v0.3.0

—— // §4 pipeline

eight workers, one shared state

step 01 planner cron · turns config into items	step 02 script-writer llm + dedup + retry	step 03 approval-gate hitl or auto	step 04 persona-picker reuse or generate
step 05 avatar-render heygen async polling	step 06 post-production ffmpeg subtitles + cta	step 07 scheduler cron-aware slot fit	step 08 publisher ig + tiktok direct

—— // §5 engineering highlights

what's interesting under the hood

[state-machine in postgres] workers claim with FOR UPDATE SKIP LOCKED – horizontal scaling, automatic crash recovery, free retries.	[hitl → autonomous] boot in hitl approval mode. one config flag flips a campaign to auto. same code path, same approval gate.
[pgvector dedup] topics embedded with text-embedding-3-small. cosine distance against last 90d. regenerates if max similarity > 0.85.	[real ffmpeg post-prod] srt subtitle burn. hook + cta overlays. tiktok-spec h.264/aac normalization. all gated by DEMO_MODE.
[multi-account routing] per-platform groups with three strategies: fan-out, round-robin, or weighted-random. tracks usage for fairness.	[analytics feedback loop] ig insights + tiktok display api → post_metrics → topic_performance → planner_weight_modifier. closed loop.

—— // §6 stack

production-credible primitives

- postgres 16 + pgvector + uuid-ossf + citext + pg_trgm – 13 states, 1536-dim embeddings, ivfflat ANN index
- typescript monorepo (pnpm workspaces) – services/core (drizzle orm, providers), services/workers, services/api (hono), dashboard (next.js 15)

- **n8n orchestration** – three importable workflows for cron, slack approval routing, publishing health monitor
- **external integrations** – openai, heygen, imagen, pexels, ig graph api, tiktok content posting – every one gated behind DEMO_MODE with a mock impl

—— // §7 quickstart

three commands to running pipeline

```
# 1. boot the database
$ cp .env.example .env && pnpm dev:db

# 2. install + seed the demo campaign
$ pnpm install && pnpm seed

# 3. run the api + workers + dashboard
$ pnpm dev:all
# → http://localhost:3000
```